Corporate Finance

Topic 4 – A

Bernardo Ricca

Departure from MM: the hidden action framework

- We now introduce asymmetric information about actions in the context of the Holmstrom and Tirole (1997) model
- This is the hidden action or moral hazard framework
- Capital structure *matters* because it affects the assets side of the firm

Credit rationing

- The model will allow us to study specific features of credit markets
- In competitive markets with symmetric information, price is the equilibrium quantity that clears markets, that is, it makes supply equal to demand
- If there is excess demand, prices increase so that supply equals demand
- In particular, there is no rationing

Credit rationing

- In credit markets, the demand for credit might exceed supply at the equilibrium interest rate
- Why do credit markets clear through quantities (credit limits) as well as prices (interest rates)?
- Why does demand exceed supply even though lenders are free to raise the price (the interest rate)?

Credit rationing

- Bester and Hellwig (1987):

  A would-be borrower is said to be rationed if he cannot obtain the loan that he wants even though he is willing to pay the interest that the lenders are asking, perhaps even a higher interest rate. In practice such credit rationing seems to be common place. Some borrowers are constrained by fixed lines of credit which they must not exceed in any circumstances, others are refused loans altogether.

Credit rationing

- Baltensperger (1978): credit rationing occurs when

  (...) some borrower's demand for credit is turned down, even if this borrower is willing to pay all the price and nonprice elements of the loan contract.

- Priced element: interest rate; nonpriced element: collateral requirement, for example

- Note that a situation in which a borrower is turned down because he/she does not have enough collateral is not classified as credit rationing

- Similarly, the exclusion of some categories from credit markets (red-lining) because of a lack of future cash flows or collateral is not credit rationing

Credit rationing

- De Mel et al. (2008):

  We use randomized grants to generate shocks to capital stock for a set of Sri Lankan microenterprises. We find the ***average real return to capital in these enterprises is 4.6%–5.3% per month, substantially higher than market interest rates****. We then examine the heterogeneity of treatment effects. Returns are found to vary with entrepreneurial ability and with household wealth, but not to vary with measures of risk aversion or uncertainty. Treatment impacts are also significantly larger for enterprises owned by males; indeed, we find no positive return in enterprises owned by females.*

Entrepreneurs (firms)

- Have a project that requires a fixed initial investment of $I > 0$ at $t = 0$
- Are risk-neutral, that is, they only care about expected returns
- Have initial wealth $A < I$

  $\Rightarrow$ they need to borrow $I - A$ to implement the project

Outside investors

- Have enough wealth to finance the entrepreneur
- Are risk-neutral, that is, they only care about expected returns
- Are competitive
- Can invest in an alternative option with a zero net return

  $\Rightarrow$ They are willing to lend as long as the net expected return from the operation is bigger or equal than zero

Discounting

- The risk-free interest rate is normalized to zero: $r_f = 0$
- Both outside investors and entrepreneurs discount any future payoff by $r = 0$

$$r = \underbrace{\text{risk premium}}_{=0 \text{ (risk neutrality)}} + r_f = 0$$

Project: financial returns and moral hazard

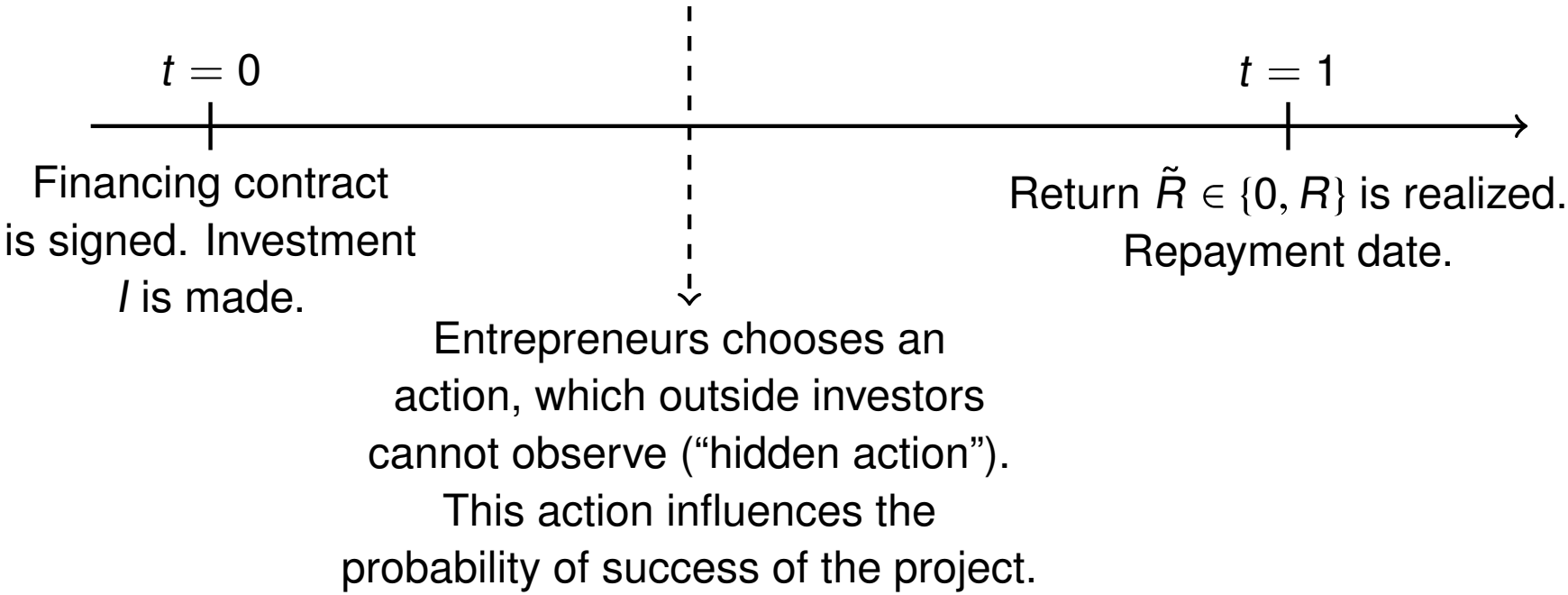
- The project yields random verifiable income $\tilde{R}$ at $t = 1$:

$$\tilde{R} = \begin{cases} R > 0, & \text{if the project is successful} \\ 0, & \text{if the project fails} \end{cases}$$

- Moral hazard: the probability of success depends on an **unobservable action** that the entrepreneur chooses between $t = 0$ and $t = 1$

- Only the entrepreneur knows about the action (private information) but all the parties know about the outcome of the project (success or failure)

Timeline


$t = 0$
$t = 1$
Financing contract is signed. Investment I is made.
Entrepreneurs chooses an action, which outside investors cannot observe ("hidden action"). This action influences the probability of success of the project.
Return $\tilde{R} \in \{0, R\}$ is realized. Repayment date.

Project: financial returns and moral hazard

- Only two possible actions: $a \in \{a_L, a_H\}$
  - Under a_H, the probability that the project succeeds is p_H
  - Under a_L, the probability that the project succeeds is p_L
- We have that $p_H > p_L$, that is, the probability of success is higher when the entrepreneur works hard
- Consequently, the NPV of the project is higher when the entrepreneur works hard:

$$\underbrace{p_H R + (1 - p_H)0 - I}_{\text{NPV when } a = a_H} = p_H R - I > p_L R - I = \underbrace{p_L R + (1 - p_L)0 - I}_{\text{NPV when } a = a_L}$$

Project: private benefits

- a_L entails a **non-pecuniary private benefit** to the entrepreneur, which is equivalent to $B > 0$ dollars

- Two equivalent interpretations of B:

  - **Effort interpretation**: effort is costly in terms of utility (working less and having more time for leisure is preferred to working hard). Therefore, B can be interpreted as the disutility of effort that is saved by shirking. Under this interpretation we use the notation $a_H = e_H$ (high effort) and $a_L = e_L$ (low effort, shirking)

  - **Project choice interpretation**: the entrepreneur chooses a project that yields a high probability of success, or one that yields a smaller probability but that he likes more because it is easier to implement, is more prestigious, delivers perks, is more fun, is easier to commit fraud and embezzlement (e.g. buying inputs at an inflated price from a friend or family)

Project: private benefits

- Jensen and Meckling (1976):

If a wholly owned firm is managed by the owner, he will make operating decisions which maximize his utility. These decisions will involve not only the benefits he derives from ***pecuniary returns*** *but also the utility generated by various* ***non-pecuniary aspects*** *of his entrepreneurial activities such as the physical appointments of the office, (...), the level of employee discipline, the kind and amount of charitable contributions, personal relations ("love", "respect", etc.) with employees, a larger than optimal computer to play with, purchase of production inputs from friends, etc.*

- While financial returns can be shared with outside investors, non-pecuniary aspects (private benefits) accrue to the entrepreneur only

Timeline

- Implicitly, there are two time periods:
  - At $t = 0$, the parties agree on a loan contract, outside investors transfer $I - A$ to the entrepreneur and the investment occurs
  - At $t = 1$, projects yields income R in case of success and 0 otherwise, and the entrepreneur repays investors
- During the execution of the project between $t = 0$ and $t = 1$ (the moral hazard or interim stage), the entrepreneur can behave and choose a_H or misbehave and choose a_L
- As outside investors cannot observe the action choice, we refer to them as uninformed investors

Contracts

- A contract between entrepreneurs and outside (uninformed) investors is described by a repayment conditional on the project outcome (success or failure)
- Entrepreneurs are protected by limited liability, that is, they cannot pay outside investors out of their own pocket
- Limited liability implies that the payment to outside investors cannot be positive in case of failure as the firm has no cash
- Therefore the payment to investors if the project fails is zero in any contract

Contracts

- The contract can then simply be described by a payment R_u to outside (uninformed) investors if the project is successful
- The (firm) entrepreneur's compensation in case of success is then $R_f = R - R_u$

	Failure	Success
Total cash flow	0	R
Outside investors	0	R_u
Entrepreneur	0	$R_f = R - R_u$

Contracts

- As outside investors cannot observe the action chosen by the entrepreneur, contracts cannot specify it
- If outside investors want the entrepreneur to choose to a_H, they must provide incentives

Contracts: one possible interpretation

- We can interpret the contract as a debt contract (loan): entrepreneur borrows $I - A$ at $t = 0$ and promises to repay R_u (the "face value") at $t = 1$
- If the project is successful, the borrower can meet the contractual obligation, but if the project fails the borrower defaults on the loan
- The interest rate i on the loan is

$$1 + i = \frac{R_u}{I - A} \iff i = \frac{R_u}{I - A} - 1$$

- For a fixed borrowed quantity $I - A$, the higher the R_u the higher the interest rate

Conflict of interests

- Entrepreneur chooses the action considering security benefits or financial returns, which are shared with outside investors, and private benefits, which accrue to the entrepreneur only $\Rightarrow$ conflict of interest

- Given a contract R_u, outside investors always want the entrepreneur to choose a_H to maximize the expected value of their investment: $p_H R_u > p_L R_u$

- Choosing a_H also maximizes the entrepreneur's expected financial return as $p_H R_f > p_L R_f$, but choosing a_L generates a private benefit B that only benefits the entrepreneur

- It is possible that

$$\underbrace{p_H R_u > p_L R_u}_{\substack{\text{Outside investors} \\ \text{prefer } a_H}} \quad \text{but} \quad \underbrace{p_H R_f < p_L R_f + B}_{\substack{\text{Entrepreneur} \\ \text{prefers } a_L}}$$

Assumptions

- **(Assumption A1)** If the entrepreneur chooses a_H, the project has a positive NPV:

$$p_H R - I > 0$$

- **(Assumption A2)** If the entrepreneur chooses a_L, the project has a negative NPV, even accounting for the private benefit gain:

$$p_L R - I + B < 0$$

Assumptions: implications

- a_L is inefficient, that is, total surplus is higher when entrepreneurs choose a_H:

$$\underbrace{(p_H - p_L)R}_{\substack{\text{gains from} \\ \text{choosing } a_H \text{ over } a_L}} - \underbrace{B}_{\substack{\text{costs from} \\ \text{choosing } a_H \text{ over } a_L}} > 0$$

Assumptions: implications

- A contract that induces the choice a_L is not viable:

$$p_L R - I + B < 0 \Rightarrow p_L \underbrace{(R_u + R_f)}_{=R} - I + B + \underbrace{A - A}_{=0} < 0$$

or

$$\underbrace{[p_L R_u - (I - A)]}_{\text{Outside investors' NPV}} + \underbrace{[p_L R_f + B - A]}_{\text{Borrower's NPV}} < 0$$

- Either the entrepreneur or the outside investors (or both) end up with a worse payoff in comparison to the situation in which they do not sign a contract, and thus the party that is worse off prefers not to sign the contract

- Bottom line: the loan contract should provide incentives for the choice a_H

Incentive constraint

- The loan contract incentivizes the choice a_H if the incentive constraint (IC) is satisfied:

$$\underbrace{p_H R_f}_{\substack{\text{utility when} \\ \text{choosing } a_H}} \geq \underbrace{p_L R_f + B}_{\substack{\text{utility when} \\ \text{choosing } a_L}}$$

or, letting $\Delta p = p_H - p_L$,

$$\underbrace{\Delta p R_f}_{\substack{\text{entrepreneur's expected financial} \\ \text{gain from choosing } a_H \text{ over } a_L}} \geq \underbrace{B}_{\substack{\text{entrepreneur's private benefit} \\ \text{loss from choosing } a_H \text{ over } a_L}}$$

Incentive constraint

- Rearranging,

$$\underbrace{R_f}_{\substack{\text{entrepreneur's compensation} \\ \text{in case of success}}} \geq \underbrace{B/\Delta p}_{\substack{\text{minimum entrepreneur's} \\ \text{compensation consistent with } a_H}}$$

- The entrepreneur must receive a sufficiently large share of the returns (enough "skin in the game") to forgo private benefits
- As $B/\Delta p$ is the minimum amount required by the entrepreneur in case of success that is consistent with a_H being optimal, the maximum amount that outside investors can receive is $R - B/\Delta p$
- We refer to $R - B/\Delta p$ as the pledgeable income

Pledgeable income

- Note that the pledgeable income $R - B/\Delta p$ does not depend on A and as a consequence on the amount borrowed $I - A$
- The maximum amount that outside investors can receive in case of success is the same if
  - A is large and thus the amount borrowed $I - A$ is small (low leverage firm)
  - A is small and thus the amount borrowed $I - A$ is large (high leverage firm)
- As lenders must break even, it will be easier to finance "rich" individuals with a large A

Individual rationality constraint

- The project is financed if outside investors breaks even. Otherwise, they prefer to obtain zero from the outside option
- Thus a necessary condition for a loan to be granted is

$$\underbrace{p_H R_u}_{\text{expected payment}} \geq \underbrace{I - A}_{\text{lender's outlay}}$$

- We refer to this condition as the individual rationality (IR) or the participation constraint (PC)

The IR and IC conditions together

- As the maximum amount that outside investors can receive in case of success is $R - B/\Delta p$, the maximum expected payment (known as expected pledgeable income) is

$$p_H\left(R - \frac{B}{\Delta p}\right)$$

- Financing takes place if and only if

$$p_H\left(R - \frac{B}{\Delta p}\right) \geq I - A$$

The IR and IC conditions together

- Thus financing takes place if and only if the entrepreneur has enough initial wealth

$$p_H\left(R - \frac{B}{\Delta p}\right) \geq I - A \iff A \geq \underbrace{p_H\left(\frac{B}{\Delta p}\right) - (p_H R - I)}_{\underline{A}}$$

- If $\underline{A} > 0$, a borrower with wealth $A < \underline{A}$ is not granted a loan
- We make the following assumption to guarantee that some borrowers do not obtain a loan
- **(Assumption A3)**: $\underline{A} = p_H\left(\frac{B}{\Delta p}\right) - (p_H R - I) > 0$

Equilibrium

- If financing takes place, that is, if $A \geq \underline{A}$, perfect competition among lenders imply zero profits:

$$p_H R_u - (I - A) = 0 \quad \text{or} \quad R_u = \frac{I - A}{p_H}$$

- If we interpret R_u as the amount due in a loan of size $I - A$, the interest rate of the contract is

$$i = \frac{R_u}{I - A} - 1 = \frac{1}{p_H} - 1$$

- Payoff of an entrepreneur with wealth $A \geq \underline{A}$ is

$$p_H(R - R_u) - A = p_H\left(R - \frac{I - A}{p_H}\right) - A = p_H R - I$$

that is, he/she receives the entire NPV (surplus)

Credit rationing

- A borrower with wealth $A < \underline{A}$ does not obtain credit
- A smaller A means that the amount borrowed, $I - A$, is larger
- Lenders must break even, so lending a larger amount means that a larger fraction of the return R must be pledged to them
- But lenders anticipate that if the fraction of R that they require increases such that the borrower receives less than $B/\Delta p$, the borrower will choose a_L and the lenders' expected payoff will be negative
- Lenders simply turn down entrepreneurs with low A. An increase in the interest rate only exacerbates the moral hazard problem (smaller payoff in case of success, more incentives to choosing a_L).

Credit rationing

- The borrower is willing to pay the equilibrium interest rate of $1/p_H - 1$ (or even a higher one)
- Entrepreneurs with low A face a commitment problem: they would be better off by exerting choosing a_H. However, once the contract is signed, they have the incentives to choose a_L
- Entrepreneurs with low A do not obtain credit and bear the moral hazard costs

Credit rationing

- Consider a borrower with $A < \underline{A}$ that promises to repay $(I - A) \times (1/p_H)$, the amount borrowed plus equilibrium interest payments
- As $A < \underline{A}$, the entrepreneur's compensation in case of success satisfies

$$R - \frac{I - A}{p_H} < R - \frac{I - \underline{A}}{p_H} = R - \frac{I - \left[p_H\left(\frac{B}{\Delta p}\right) - (p_H R - I)\right]}{p_H} = \frac{B}{\Delta p}$$

- As the compensation is smaller than the minimum required to induce a_H, the entrepreneur ends up choosing a_L, leaving the lender with negative profits

Determinants of credit rationing

- $\underline{A}$ is increasing in B
- $\underline{A}$ decreases in $\Delta p/p_H$
  - $\Delta p/p_H$ is a measure of the marginal gain of choosing a_H over a_L
  - The higher the ratio $\Delta p/p_H$, the more informative is the outcome about the action choice

Efficiency of the equilibrium

- The allocation is inefficient:
  - Recall that choosing a_H is efficient (it increases total surplus)
  - With no asymmetric information about the action chosen, there is no need for an incentive constraint: a_H can simply be specified in the contract
  - In the competitive equilibrium (first-best) all entrepreneurs obtain financing and exert a_H
  - In particular, there is no credit rationing

Alternative interpretation of A

- We can interpret A as collateral instead of initial wealth when liquidation costs are zero
- In this case, entrepreneurs post A as collateral and borrow I
- The payoff of outside investors is $p_H R_u + (1 - p_H)A$ and the IC constraint becomes

$$p_H(R - R_u) + (1 - p_H)(-A) \geq p_L(R - R_u) + (1 - p_L)(-A) + B$$

- The cutoff $\underline{A}$ would be the same

Sensibility of investment to cash flows

- Credit rationing is a potential explanation for the fact that investment is sensitive to cash flow
- Fazzari and Petersen (1988):
  - Robust relationship between cash flows and investment
  - Larger for more “constrained” firms
- Empirical challenge: finding an exogenous increase in cash flows or cash balances (e.g. De Mel et al., 2008)

Sensibility of investment to cash flows

- In our model: a small increase in cash flows increase investment for marginal firms, the ones with initial cash balances just below $\bar{A}$
- No effect for “very small” or “very large” firms
- Credit rationing makes the marginal value of an increase in cash flows larger than 1:
  - An additional unit of cash increases the entrepreneur’s utility by more than one if this additional unit allows the entrepreneur to access credit markets

Debt versus equity

- To finance the purchase of fixed assets and inputs (raw materials, wages) that enable the sale of services and products, firms can issue
  - **Liabilities** or **promises to non-owners (creditors)**: specific and senior (e.g. bonds, bank loans, payables)
  - **Equity** or **promises to owners (shareholders)**: non-specific and junior (e.g. stocks)
- Owners are residual claimants (their claim is junior to liabilities): they receive whatever future cash flows that are not promised away to non-owners (creditors)
- In other words, creditors have priority (their claim is senior to equity), and shareholders only get paid after fulfilling the promises to creditors

Standard debt contracts

- A standard debt contract is one of the most common forms of liability
- Does this contract have good properties in terms of facilitating access to funding in the presence of moral hazard?
- Key characteristics:
  - The contract specifies fixed payments to the lender at fixed times
  - If the borrower misses a payment (defaults), the lender can seize the firm's assets, leaving equityholders empty-handed
  - Under limited liability, only assets of the firm can be seized
  - In some cases, in case of default, banks can also report negative information about the borrower to a credit bureau

Payoffs: equity vs debt

- Consider a firm (entrepreneur) that finances its operations with equity and a one-period debt contract that specifies a fixed repayment of D to creditors at maturity

- If the firm has assets A at maturity, we have two outcomes:

  - If $A < D$, the firm defaults on its debt. Creditors are senior and seize all the assets, receiving A; equityholders are empty-handed

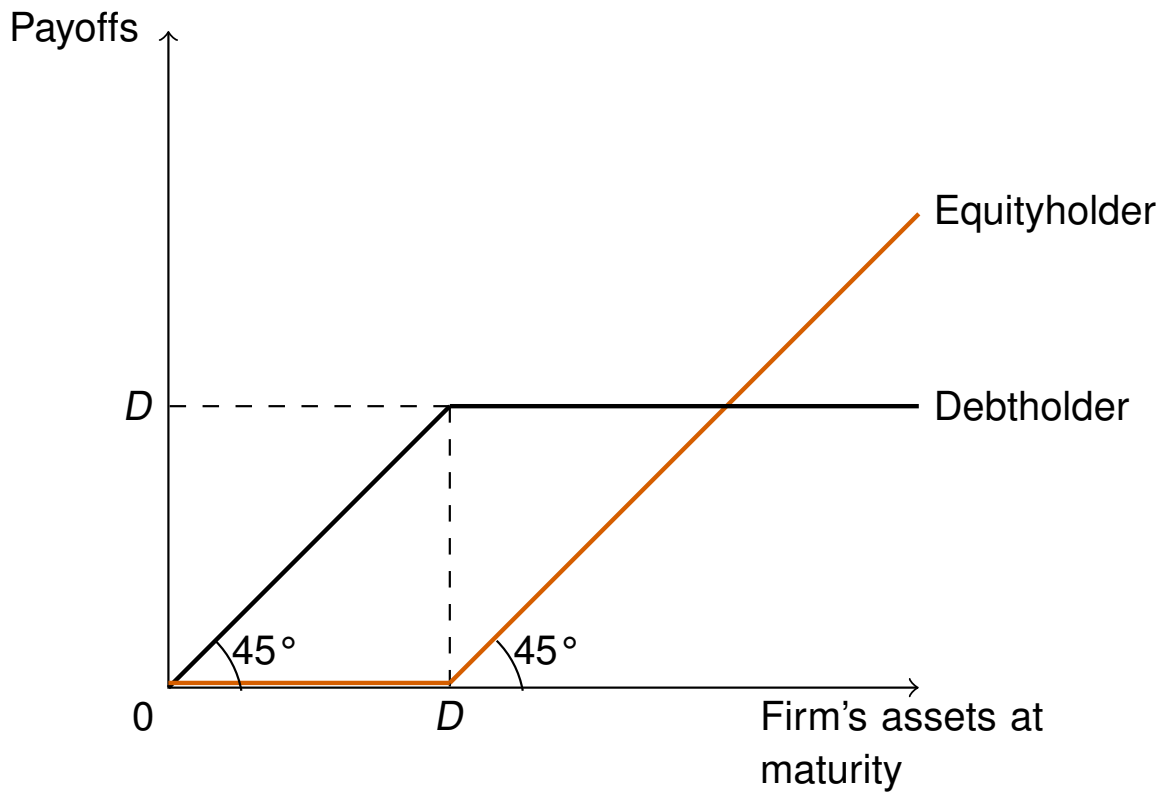
  - If $A \geq D$, the firm can fulfill its promise to repay the fixed amount D; the equity value is the residual $A - D$

- The payoffs of debtholders and equityholders are (as a function of A)

	Default $A < D$	Full repayment $A \geq D$
Debt	A	D
Equity	0	$A - D$

Payoffs: equity vs debt

- Payoffs at maturity for a debt contract with no coupon payments and face value D



Direct financing: debt or equity?

- Payment to outside investors:

$$\begin{cases} R_u = \frac{I-A}{p_H}, & \text{if the project is succeeds} \\ 0, & \text{if the project fails} \end{cases}$$

- The contract can be interpreted as debt contract: in exchange for receiving $I - A$, entrepreneurs promise to repay a fixed amount (face value) of $D = R_u$

	Failure (default)	Success (repayment)
Total cash flow (asset)	0	R
Debt (outside investors)	0	R_u
Equity (entrepreneurs)	0	$R - R_u$

Direct financing: debt or equity?

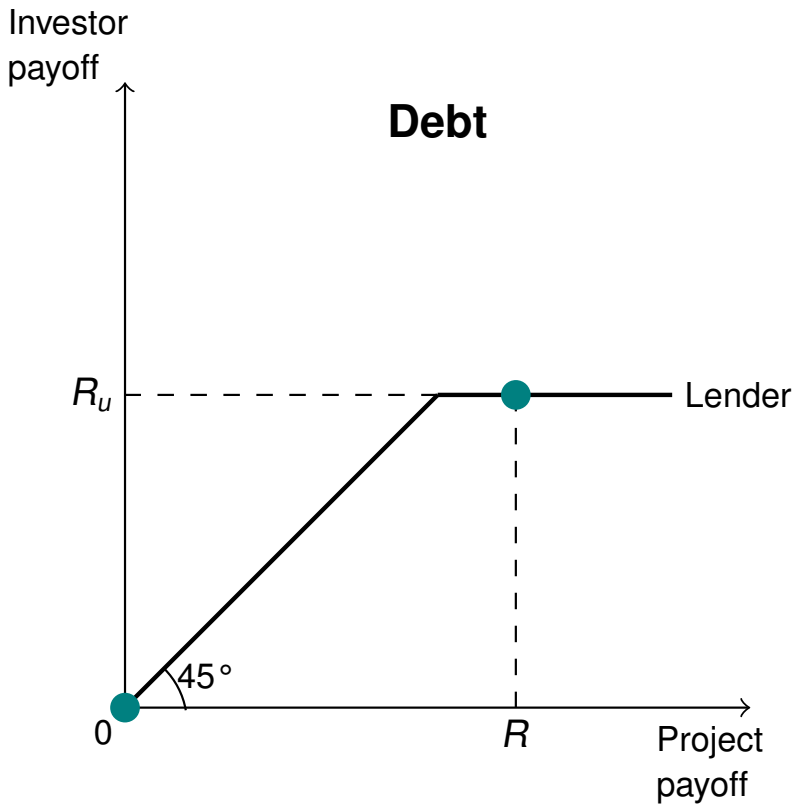
- The contract can also be interpreted as equity: in exchange for receiving $I - A$, entrepreneurs sell a share $\alpha = R_u/R$ of the firm to outside investors

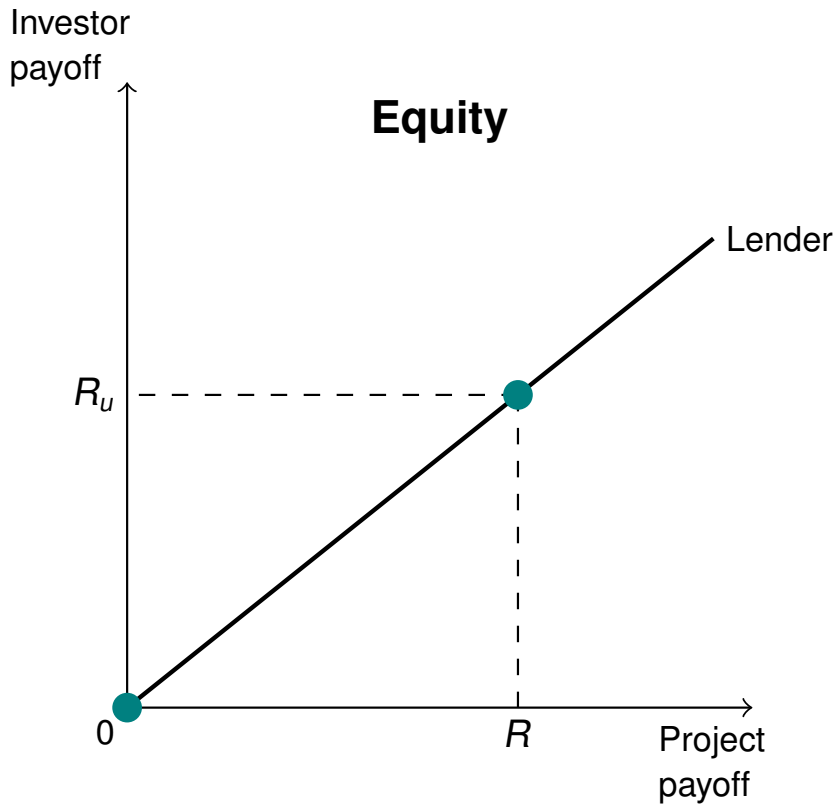
	Failure	Success
Total cash flow (asset)	0	R
Equity (outside investors)	$\alpha 0 = 0$	$\alpha R = R_u$
Equity (entrepreneurs)	$(1-\alpha)0 = 0$	$(1-\alpha)R = R - R_u$

- This multiplicity arises because we only have two states (assets can take two values only), and in one of them the project yields a zero return

- In general, the payoffs of debt and equity securities are different. As a consequence they can differ in terms of incentives provision and in the mitigation of the hazard problem

Debt versus equity





Project: new payoffs

- To gain insights about security design and the differences between debt and equity, assume that the project has a positive payoff in both states:

$$\tilde{R} = \begin{cases} R^S, & \text{if the project succeeds} \\ R^F, & \text{if the project fails} \end{cases}$$

  where $R^S > R^F > 0$

- The initial investment remains I, where $R^F < I < R^S$

Contracts

- We assume limited liability
- Contract: outside investors receive $R_u^S \in [0, R^S]$ in case of success and $R_u^F \in [0, R^F]$ in case of failure
- The entrepreneur receives R_f^S in case of success and R_f^F in case of failure

	Failure	Success
Total cash flow	R^S	R^F
Outside investors	R_u^S	R_u^F
Entrepreneur	R_f^S	R_f^F

- $R^S = R_u^S + R_f^S$ and $R^F = R_u^F + R_f^F$

Assumptions

- **(A1)** Positive NPV if the worker behaves:

$$p_H R^S + (1 - p_H) R^F - I > 0$$

- **(A2)** Negative NPV if the worker shirks (even if B is taken into account):

$$p_L R^S + (1 - p_L) R^F + B - I < 0$$

- Again, given (A2), the loan contract should provide incentives for the action a_H, otherwise one of the parties is better off by not participating in the transaction

First-best

- No asymmetric information: the action choice is observable and verifiable
- There is no need to incentivize the entrepreneur (no need for an incentive constraint): the contract can simply specify a_H
- Modigliani-Miller result: financing choices are irrelevant in the absence of moral hazard
  - The entrepreneur can raise $I - A$ by issuing debt
  - The entrepreneur can raise $I - A$ by issuing equity (selling part of the firm)
  - Or any combination of debt and equity (say, the entrepreneur can raise half of $I - A$ by issuing debt and the other half by issuing equity)

First-best: implementation with a debt contract

- Assume that $I - A > R^F$ (otherwise the entrepreneur can issue riskless debt)

- **Risky debt** with face value $D > R^F$ such that the payoff to outside investors satisfies:

$$R_u = \begin{cases} D, & \text{if the project succeeds (full repayment)} \\ R^F, & \text{if the project fails (default)} \end{cases}$$

where D is such that

$$p_H D + (1 - p_H) R^F = I - A \;\Rightarrow\; D = R^F + \frac{I - A - R^F}{p_H}$$

First-best: implementation with an equity contract

- **Equity**: to raise $I - A$, the entrepreneur sells a share α of the project to outside investors:

$$R_u = \begin{cases} \alpha R^S, & \text{if the project succeeds} \\ \alpha R^F, & \text{if the project fails} \end{cases}$$

where α is such that

$$\alpha[p_H R^S + (1 - p_H) R^F] = I - A \Rightarrow \alpha = \frac{I - A}{p_H R^S + (1 - p_H) R^F}$$

- Irrespective of the financing method (which can include arbitrary combinations of debt and equity), the project is always financed and the entrepreneur gets the entire NPV

Asymmetric information: action is not observable

- Financing takes place if and only if

$$\text{expected pledgeable income} \geq \underbrace{I - A}_{\text{amount lent / invested}}$$

  where the expected pleadgeble income is the maximum expected amount that lenders can receive that is consistent with entrepreneurs having incentives to choose a_H

- The higher the pledgeable income, the larger the amount that outside investors can lend/invest and the smaller the credit rationing problem:
  - More entrepreneurs (with smaller initial wealth A) can obtain credit

- Which contract – debt or equity – maximizes pledgeable income?

Asymmetric information: action is not observable

- The entrepreneur chooses a_H only if such a choice generates more utility than shirking

- The incentive constraint is

$$p_H R_f^S + (1 - p_H) R_f^F > p_L R_f^S + (1 - p_L) R_f^F + B$$

or

$$R_f^S - R_f^F \geq \frac{B}{\Delta p}$$

- Note the cost of rewarding the entrepreneur in case of failure:
  - A higher reward in case of failure reduces the marginal benefit of a_H
  - The keep the same marginal benefit of a_H we also need to increase the reward in case of success, decreasing the amount that lenders can obtain (i.e. decreasing the pledgeable income)

Asymmetric information: action is not observable

- We will see that the solution that maximizes pledgeable income and minimizes rationing involves $R_f^F = 0$
- Setting $R_f^F = 0$ minimizes the costs to provide incentives and thus maximizes the amount that lenders can receive

The choice that maximizes pledgeable income

- Given the IC constraint, the expected pledgeable income or the maximum expected payoff of outside investors is

$$
\begin{aligned}
& p_H R_u^S + (1 - p_H) R_u^F \\
&= p_H (R^S - R_f^S) + (1 - p_H)(R^F - R_f^F) \\
&= p_H (R^S - R^F) + R^F - p_H \underbrace{(R_f^S - R_f^F)}_{\text{max. } B/\Delta p} - R_f^F \\
&= p_H (R^S - R^F) + R^F - p_H \frac{B}{\Delta p} - \underbrace{R_f^F}_{\substack{\text{the choice variable} \\ \text{that determines the} \\ \text{max. pledgeable income}}}
\end{aligned}
$$

- Setting $R_f^F = 0$ (leaving the entrepreneur empty-handed in case of failure) maximizes the outside investors' expected payoff

The choice that maximizes pledgeable income

- How can we obtain $R_f^F = 0$?
- The financing choice must contain debt with face value $D \geq R^F$
- Otherwise, the entrepreneur will have a positive payoff in case of failure
- Leaving the entrepreneur empty-handed in case of failure can be interpreted as a “punishment”, which is more likely to occur when the entrepreneur chooses a_L

Suboptimality of all-equity financing

- All-equity financing is sub-optimal because the borrower gets a positive payoff in case of failure $\Rightarrow$ positive reward for a_L
- Denote by α the share of the project that entrepreneurs sell to outside investors, meaning that entrepreneurs retain a share $1-\alpha$ of the firm
- The IC constraint is

$$(1-\alpha)\underbrace{\left[p_H R^S + (1-p_H)R^F\right]}_{\text{firm value when } a = a_H} > (1-\alpha)\underbrace{\left[p_L R^S + (1-p_L)R^F\right]}_{\text{firm value when } a = a_L} + B$$

or

$$1-\alpha \geq \frac{B}{\Delta p(R^S - R^F)}$$

Suboptimality of all-equity financing

- In case of failure, the entrepreneur gets at least

$$\underbrace{\frac{B}{\Delta p(R^S - R^F)}}_{1-\alpha} \times R^F \quad > \quad \underbrace{0}_{\substack{\text{payoff under} \\ \text{debt with } D > R^F}}$$

- The maximum share of the firm that outside investors can have is

$$1 - \frac{B}{\Delta p(R^S - R^F)}$$

Suboptimality of all-equity financing

- Financing takes place if the expected pledgeable income is bigger than $I - A$, that is, if outside investors can break even:

$$\left(1 - \frac{B}{\Delta p(R^S - R^F)}\right)\left[p_H R^S + (1 - p_H)R^F\right] \geq I - A$$

- Rearranging, the minimum welath required to obtain all-equity financing is

$$\underline{A}_{all-equity} = \frac{BR^F}{\Delta p(R^S - R^F)} + p_H \frac{B}{\Delta p} - \left(p_H(R^S - R^F) + R^F - I\right)$$

All-debt financing

- Consider a debt contract with face value $D = R_u^S > R^F$
- Then, in case of failure (default), the payoffs are $R_f^F = 0$ and $R_u^F = R^F$
- In case of success, the minimum compensation R_f^S of the entrepreneur is $B/\Delta p$ and thus the maximum compensation R_u^S of outside investors, the pledgeable income, is $R^S - B/\Delta p$
- Therefore expected pledgeable income is

$$p_H\Big(R^S - \frac{B}{\Delta p}\Big) + (1 - p_H)R^F$$

All-debt financing

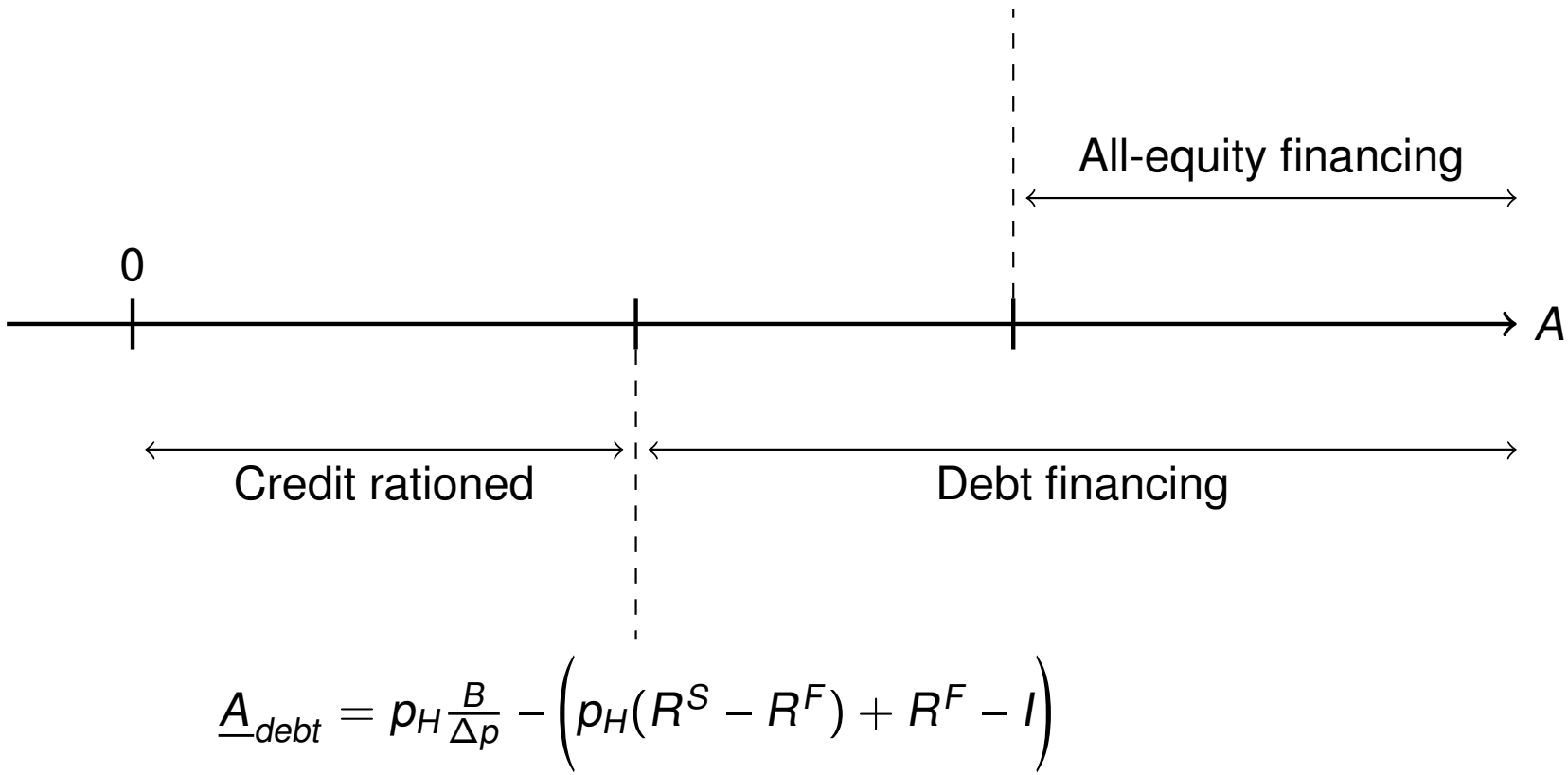
- Financing takes place if the expected pledgeable income is bigger than $I - A$, that is, if outside investors can break even:

$$p_H\Big(R^S - B\Delta p\Big) + (1 - p_H)R^F \geq I - A$$

- Rearranging, the minimum welath required to obtain debt financing is

$$\underline{A}_{debt} = p_H \frac{B}{\Delta p} - \Big(p_H(R^S - R^F) + R^F - I\Big)$$

$$\underline{A}_{all-equity} = \frac{BR^F}{\Delta p(R^S - R^F)} + p_H \frac{B}{\Delta p} - \left(p_H(R^S - R^F) + R^F - I \right)$$



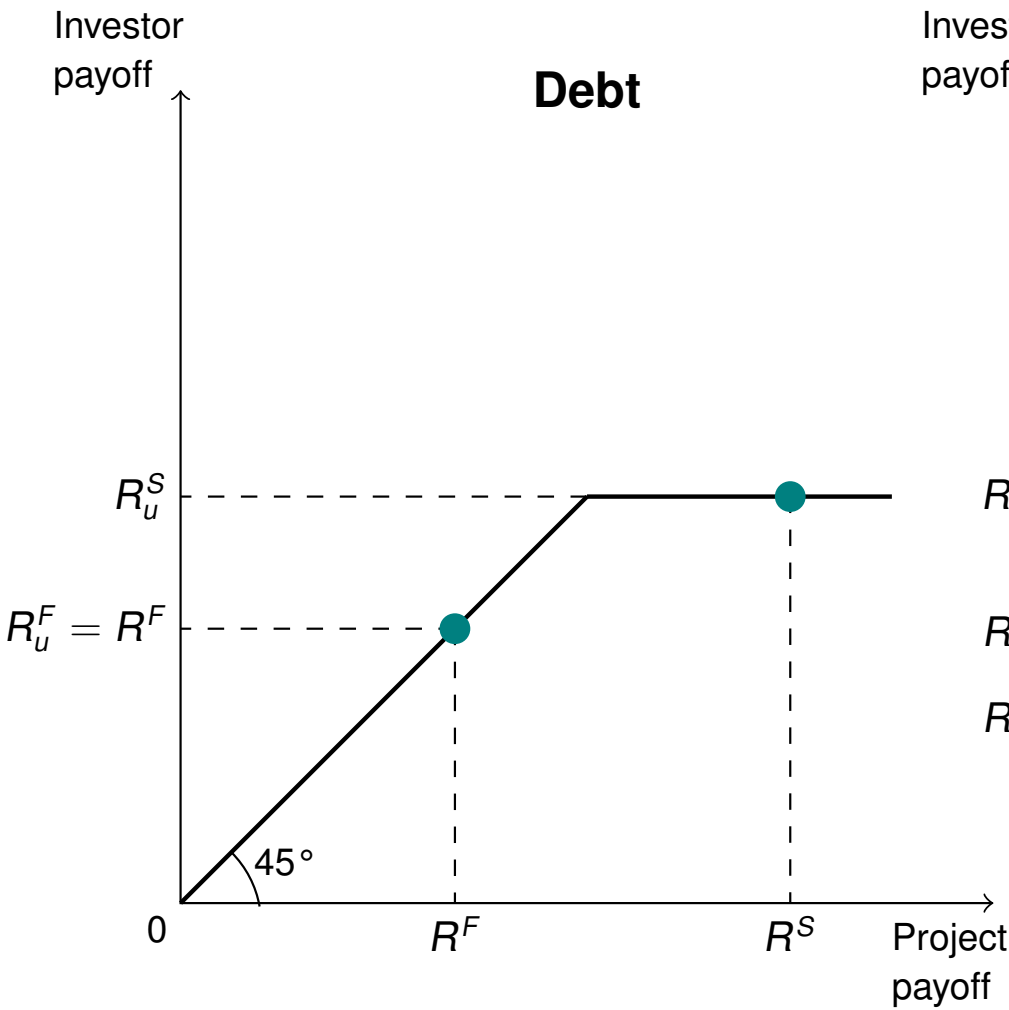
$$\underline{A}_{debt} = p_H \frac{B}{\Delta p} - \left(p_H(R^S - R^F) + R^F - I \right)$$

- Projects of individuals with initial wealth $\underline{A}_{debt} \leq A < \underline{A}_{all-equity}$ can be financed with debt with face value $D \geq R^F$, but cannot be financed with equity only $\Rightarrow$ debt dominates equity

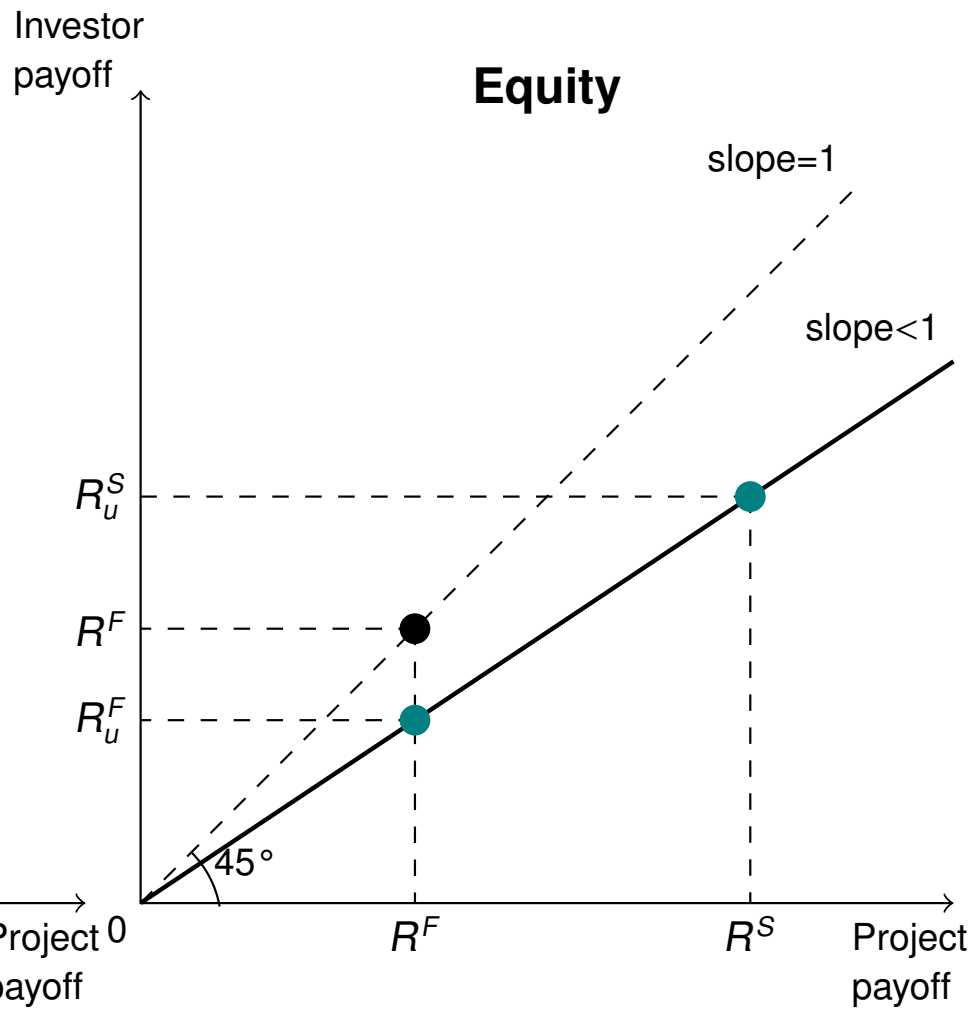
Debt vs equity

- Setting $R_f^F = 0$ maximizes investors' payoff and thus makes the participation constraint of the lender the least severe
- All projects that can be financed with equity can also be financed with debt, but the reverse does not hold
- Intuition: debt contract maximizes the return from choosing a_H that accrues to the entrepreneur
- Solution is not unique. The financial structure can involve debt only, with face value $D > R^F$, or a combination of riskless debt with face value $D = R^F$ and some equity. In any case, entrepreneurs must receive zero when the project fails

Debt versus equity



$$R_u^F = R^F \Rightarrow R_f^F = 0$$



$$R_u^F < R^F \Rightarrow R_f^F > 0$$

Optimality of the debt contract: limitations

- Debt is not a panacea
- Modigliani-Miller: the financing method is irrelevant
- Optimality arises in this specific moral hazard setting with the assumption of risk-neutrality: high-powered incentives are optimal because there is no need for risk-sharing (and consumption smoothing)
- If the entrepreneur is risk-averse, the entrepreneur bears too much risk with a debt contract

Optimality of the debt contract: limitations

- Leverage also increases the risk of bankruptcy or financial distress
- Financial distress and bankruptcy have costs that drain resources from the firm and distort managerial decisions
- **Direct bankruptcy costs**: legal and administrative costs, time and resources spent by managers and creditors dealing with bankruptcy, fire sales of assets due to asset specificity and illiquid assets
- **Indirect bankruptcy costs**: loss of customers, key employees, and access to credit markets, conflict of interest between equityholders and creditors might lead to poor investment decisions (debt overhang, asset substitution)

References I

Baltensperger, E. (1978). Credit rationing: issues and questions. *Journal of Money, Credit and Banking*, 10(2):170–183.

Bester, H. and Hellwig, M. (1987). Moral hazard and equilibrium credit rationing: An overview of the issues. In *Agency theory, information, and incentives*, pages 135–166. Springer.

Dasgupta, A. (2015). Financial economics. *Lecture Notes, LSE*.

De Mel, S., McKenzie, D., and Woodruff, C. (2008). Returns to capital in microenterprises: evidence from a field experiment. *The quarterly journal of Economics*, 123(4):1329–1372.

Fazzari, S. M. and Petersen, B. C. (1988). Financing constraints and corporate investment. *Brookings Papers on Economic Activity*, 1988(1):141–195.

Holmstrom, B. and Tirole, J. (1997). Financial intermediation, loanable funds, and the real sector. *the Quarterly Journal of economics*, 112(3):663–691.

References II

Jensen, M. C. and Meckling, W. H. (1976). Theory of the firm: Managerial behavior, agency costs and ownership structure. *Journal of financial economics*, 3(4):305–360.